RRC-Raptor-V4H SCIF/HSCIF

- User's Manual: Software

V0.0.2

Contents

LIST OF FIGURES

i. VERSION LIST

Description	Version	Date	Author
RTX_SCIF/HSCIF	V0.0.1	2024/07/22	Jerry
RTX_SCIF/HSCIF	V0.0.2	2024/10/29	Jerry

ii. HASHTAG

#V4H
#UART
#SCIF
#HSCIF
#BSP

1. Overview

1.1. Overview

Target：This manual explains the driver module that controls the SCIF and HSCIF on R-Car.

1.2. Function

This module controls the Serial Communication Interface with FIFO (SCIF) on R-Car . This module supports the following functions.

- Data transmission and reception for RS232C
- Control of communication settings
  – Baud rate 9600, 19200, 38400, 57600, 115200[bps]
  – Parity bit (none/ODD/EVEN)
  – Stop bit (1bit or 2bit)
  – Data transfer bit length (7bit or 8bit)
  Each setting can be changed from the standa
- Hardware Flow Control

– Support:
SCIF0, SCIF1, SCIF3, SCIF4
– Not support: SCIF2, SCIF5

This module controls the High Speed Serial Communication Interface with FIFO (HSCIF) on R-Car. This module supports the following functions.

- Data transmission and reception for RS232C
- Control of communication settings

– Baud rate 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600, and 1000000 ～ 3000000(Max)[bps]
– Parity bit (none/ODD/EVEN)
– Stop bit (1bit or 2bit)
– Data transfer bit length (7bit or 8bit)
Each setting can be changed from the standard ioctl for tty.

- Hardware Flow Control

2. Directory Configuration

The directory configuration is shown below.

```
────── drivers/tty/serial ────── sh-sci.c    : serial source file
                 └────────────── sh-sci.h    : serial header file
```

Figure 1. Directory Configuration

3. Enable Hardware Flow Control

The following description explain how to enable hardware flow control function. Add "uart-has-rtscts" property to use the hardware flow control function. The following description indicates a difference from the default setting. (“+”: Setting after the modification)

```
&hscif3 {
    pinctrl-0 = <&hscif3_pins>;
    pinctrl-names = "default";

    uart-has-rtscts;
    /* Please use exclusively to the scif0 node */
    status = "okay";
};
```

Figure 2. Enable flow control

4. Kernel Configuration Setting

To enable the functions of this module, make the following setting with Kernel Configuration.

```
General setup    --->
        [ ] Embedded system *1

Device Drivers    --->
        Character devices    --->
                Serial drivers    --->
                        <*> SuperH SCI(F) serial port support
                        (11)    Maximum number of SCI(F) serial ports *2
                        [*]     Support for console on SuperH SCI(F) *2
                        [*]     Support for early console on SuperH SCI(F) *2
                        [*]     DMA support *2 *3
```

Figure 3 Configuration Setting

5. Connect

提供 UART interface 讓 User 可以自己加入 UART device.

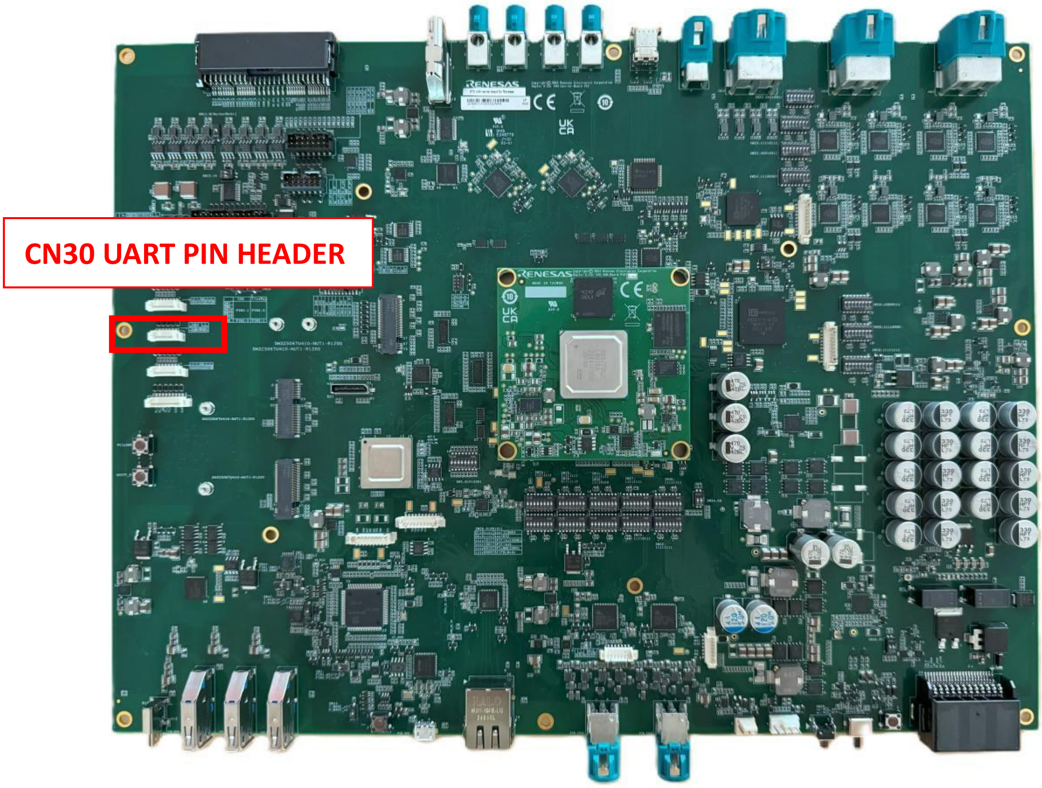

Figure 4 Connect

CN30 UART PIN HEADER

PIN	SYMBOL
1	D3.3V_SPI
2	GP1_27/HSCIF3_HCTS3N_V (**CTS**)
3	GP1_26/HSCIF3_HRTS3N_V (**RTS**)
4	GND
5	GP1_24/SL_SW1_V (**RX**)
6	GP1_28/HSCIF3_HTX3_V (**TX**)

6. UART Tool

Use stty can set UART characteristics .

stty -F /dev/ttySC3 115200 cs8 -cstopb -parenb

//Set the baudrate 115200, 8 bits, 1 stop bis, no parity

stty -F /dev/ttySC3 –crtscts

//Stop flow control

stty -F /dev/ttySC3 crtscts

//Set flow control

7. APPENDIX